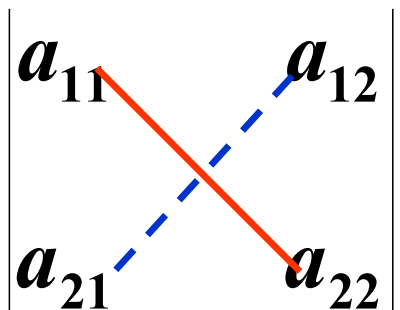


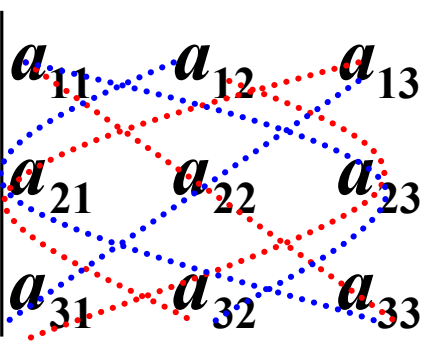


问题：什么是 n 阶行列式？

$$D = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} = ?$$

回忆


$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$


$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} \\ - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31}$$


$$D = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} = ?$$

特点: 1. n^2 个元素

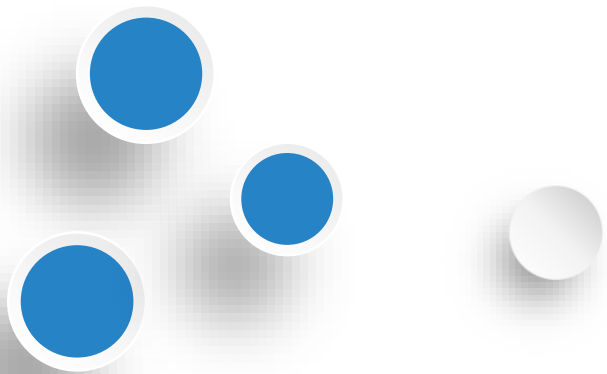
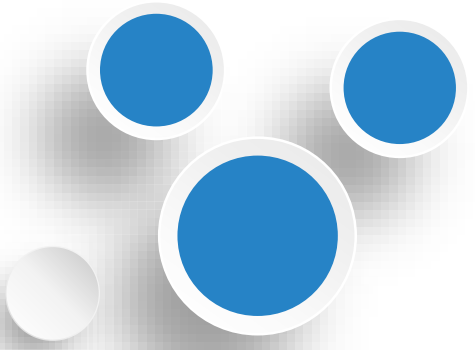
2. 共有 $n!$ 项代数和

3. 每项为取自不同行不同列的元素乘积

4. 正负项的个数相等

5. ?, 取正号

?, 取负号



§ 2

排列

§2 全排列与逆序数



1.排列的相关概念

2.排列的奇偶性

1.排列的相关概念



n 级排列: 由 $1, 2, \dots, n$ 组成的一个有序数组.

自然排列 (标准次序): $123\dots n$

逆序: 在一个排列中, 如果一对数的前后位置与大小顺序相反

逆序数: 排列中逆序的总数.

(1)

3	2	5	1	4
↓	↓	↓	↓	↓
0	1	0	3	1

于是排列32514的逆序数为

$$t = 0 + 1 + 0 + 3 + 1 = 5.$$


$$(2) \quad n(n-1)(n-2)\cdots 321$$

解

$$n(n-1)(n-2)\cdots 321$$

$$t = 1 + 2 + \dots + (n-2) + (n-1)$$

$$= \frac{n(n-1)}{2},$$

2.排列的奇偶性



奇排列：逆序数为奇数的排列

偶排列：逆序数为偶数的排列

对换： $p_1 \cdots p_i \cdots p_j \cdots p_n \longrightarrow p_1 \cdots p_j \cdots p_i \cdots p_n$

相邻对换： $p_1 \cdots p_i p_{i+1} \cdots p_n \longrightarrow p_1 \cdots p_{i+1} p_i \cdots p_n$



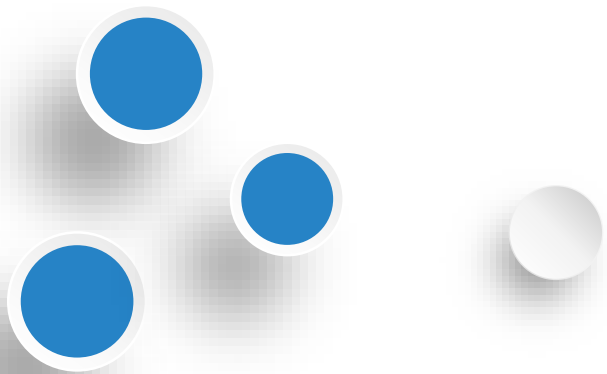
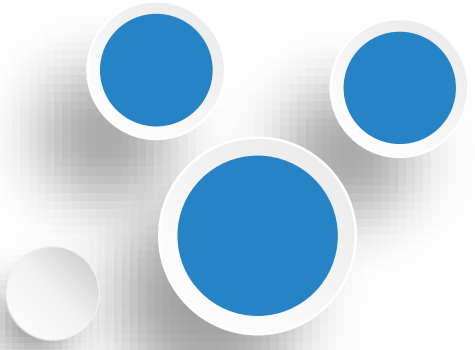
定理1：对排列进行一次对换则改变排列的奇偶性.



定理1: 对排列进行一次对换则改变排列的奇偶性

推论1: 奇（偶）排列调为自然排列所需的对换次数
为奇（偶）数

推论2: 所有 n 元排列（ $n>1$ ）中奇偶排列各半



§3

n 阶行列式的定义



一、 n 阶行列式的定义

二、特殊的几类行列式

三、 n 阶行列式的等价定义

一、 n 阶行列式的定义



$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21} = \sum_{p_1 p_2 \text{ 为 2 元排列}} (-1)^{\tau(p_1 p_2)} a_{1p_1} a_{2p_2}$$

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} \\ - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31} \\ = \sum_{p_1 p_2 p_3 \text{ 为 3 元排列}} (-1)^{\tau(p_1 p_2 p_3)} a_{1p_1} a_{2p_2} a_{3p_3}$$


$$\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} = \sum_{p_1 p_2 \dots p_n \text{ 为 } n \text{ 元排列}} (-1)^{\tau(p_1 p_2 \dots p_n)} a_{1p_1} a_{2p_2} \cdots a_{np_n}$$

特点：1. n^2 个元素

2. 共有 $n!$ 项代数和

3. 每项为取自不同行不同列的元素乘积

4. 正负项的个数相等

5. 当下标排列为偶排列时，取正号

当下标排列为奇排列时，取负号

定义 n 阶行列式

$$\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}$$

等于所有取自不同行不同列的 n 个

元素的乘积 $a_{1p_1} a_{2p_2} \cdots a_{np_n}$ (*) 的代数和,

其中 $p_1 p_2 \cdots p_n$ 是 $1, 2, \cdots, n$ 的一个排列, 每一项 (*) 都按下列规则带有符号:

当 $p_1 p_2 \cdots p_n$ 是偶排列时, (*) 带有正号,

当 $p_1 p_2 \cdots p_n$ 是奇排列时, (*) 带有负号.

n 阶行列式的定义可以写成:

$$\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} = \sum_{p_1 p_2 \cdots p_n \text{ 为 } n \text{ 元排列}} (-1)^{\tau(p_1 p_2 \cdots p_n)} a_{1p_1} a_{2p_2} \cdots a_{np_n}$$

其中 $\sum_{p_1 p_2 \cdots p_n}$ 表示对所有 n 阶排列求和,

$\tau(p_1 p_2 \cdots p_n)$ 表示对所有 n 阶排列 $p_1 p_2 \cdots p_n$ 的逆序数.

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} \quad \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} = |A| = \det A$$

思考1：矩阵与行列式的关系？

思考2：一阶行列式？



此题未设置答案，请点击右侧设置按钮

写出4阶行列式中带 $a_{12}a_{34}$ 的项

A

$$a_{12}a_{21}a_{34}a_{43}$$

B

$$a_{12}a_{23}a_{34}a_{41}$$

C

$$-a_{12}a_{21}a_{34}a_{43}$$

D

$$-a_{12}a_{23}a_{34}a_{41}$$

提交



21



二、特殊的几类行列式



例1

证明 n 阶行列式

对角形行列式

$$\begin{vmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{vmatrix} = \lambda_1 \lambda_2 \cdots \lambda_n;$$

其中未写出的元素都是0.

此题未设置答案，请点击右侧设置按钮

$$\begin{vmatrix} & & & \lambda_1 \\ & & \lambda_2 & \\ & \ddots & & \\ \lambda_n & & & \end{vmatrix} =$$

- A $\lambda_1 \lambda_2 \cdots \lambda_n$
- B $-\lambda_1 \lambda_2 \cdots \lambda_n$
- C $(-1)^{\frac{n(n-1)}{2}} \lambda_1 \lambda_2 \cdots \lambda_n,$
- D 符号无法确定

例2 计算上三角形行列式

当 $i > j$ 时, $a_{ij} = 0$

$$\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{vmatrix} = a_{11} a_{22} \cdots a_{nn}.$$

例2 计算下三角形行列式

下三角形行列式 $D = \begin{vmatrix} a_{11} & & & 0 \\ a_{21} & a_{22} & & \\ \vdots & \vdots & \ddots & \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} = a_{11} a_{22} \cdots a_{nn}.$

当 $i < j$ 时, $a_{ij} = 0$



例3 证明

$$\begin{vmatrix} a_{11} & \mathbf{0} & \cdots & \mathbf{0} \\ a_{21} & b_{11} & \cdots & b_{1n-1} \\ \vdots & \vdots & & \vdots \\ a_{n1} & b_{n-11} & \cdots & b_{n-1n-1} \end{vmatrix} = a_{11} \begin{vmatrix} b_{11} & \cdots & b_{1n-1} \\ \vdots & \vdots & \vdots \\ b_{n-11} & \cdots & b_{n-1n-1} \end{vmatrix}$$

三、 n 阶行列式的等价定义

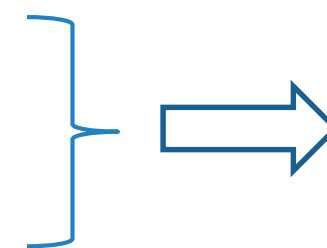


$$D = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} \stackrel{?}{=} \sum_{q_1 q_2 \cdots q_n} (-1)^{\tau(q_1 q_2 \cdots q_n)} a_{q_1 1} a_{q_2 2} \cdots a_{q_n n}$$
$$= \sum_{p_1 p_2 \cdots p_n} (-1)^{\tau(p_1 p_2 \cdots p_n)} a_{1 p_1} a_{2 p_2} \cdots a_{n p_n}$$

$$a_{1p_1} a_{2p_2} \cdots a_{np_n} \longrightarrow a_{q_1 1} a_{q_2 2} \cdots a_{q_n n}$$

$$p_1 p_2 \cdots p_n \xrightarrow{s\text{次}} 12 \cdots n \quad (-1)^{\tau(p_1 p_2 \cdots p_n)} = (-1)^s$$

$$12 \cdots n \xrightarrow{s\text{次}} q_1 q_2 \cdots q_n \quad (-1)^{\tau(q_1 q_2 \cdots q_n)} = (-1)^s$$



$$(-1)^{\tau(p_1 p_2 \cdots p_n)} a_{1p_1} a_{2p_2} \cdots a_{np_n} = (-1)^{\tau(q_1 q_2 \cdots q_n)} a_{q_1 1} a_{q_2 2} \cdots a_{q_n n}$$

三、n阶行列式的等价定义

$$D = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} = \sum_{q_1 q_2 \cdots q_n} (-1)^{\tau(q_1 q_2 \cdots q_n)} a_{q_1 1} a_{q_2 2} \cdots a_{q_n n}$$
$$= \sum_{q_1 q_2 \cdots q_n} (-1)^{\tau(q_1 q_2 \cdots q_n) + \tau(k_1 k_2 \cdots k_n)} a_{q_1 k_1} a_{q_2 k_2} \cdots a_{q_n k_n}$$

课后自行证明

思考题

已知 $f(x) = \begin{vmatrix} x & 1 & 1 & 2 \\ 1 & x & 1 & -1 \\ 3 & 2 & x & 1 \\ 1 & 1 & 2x & 1 \end{vmatrix}$

求 x^3 的系数.

解 含 x^3 的项有两项,即

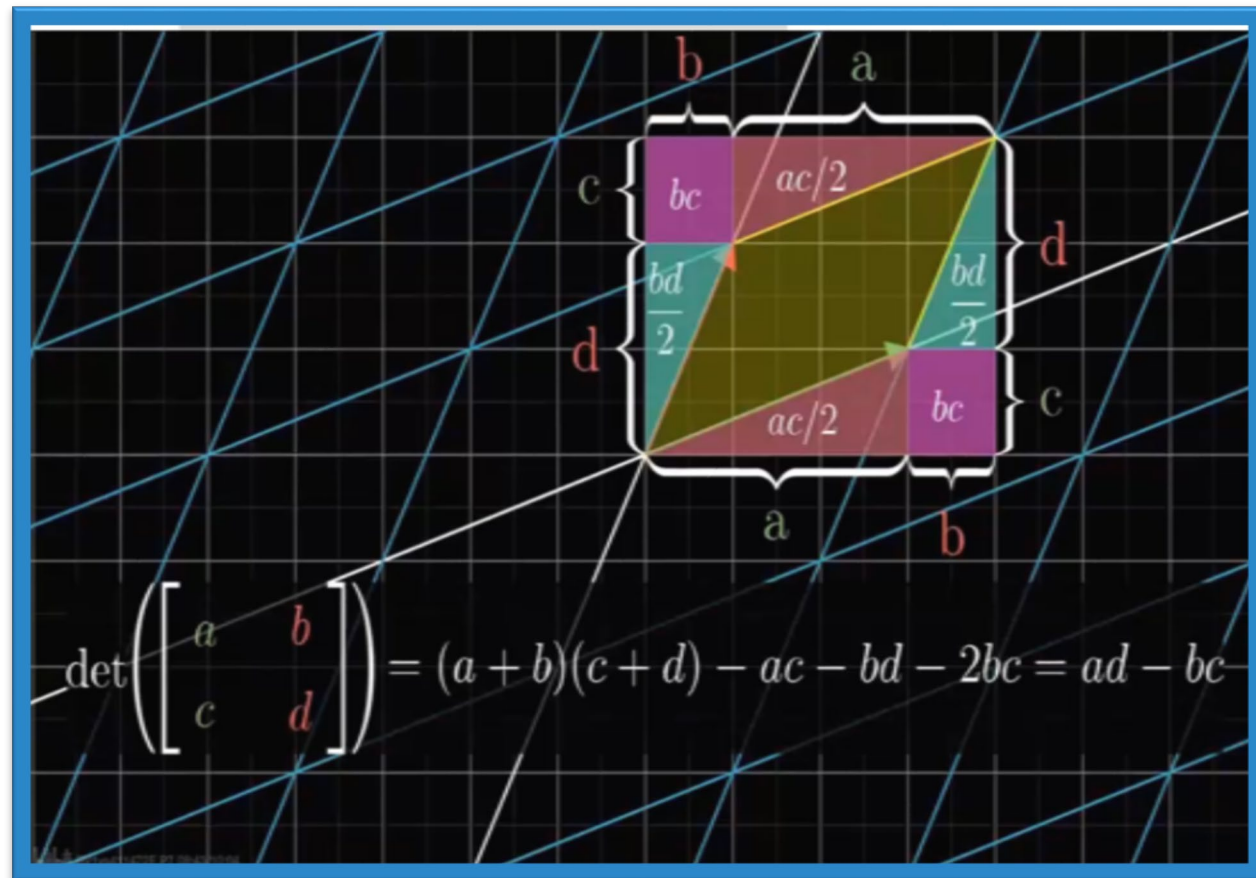
$$f(x) = \begin{vmatrix} x & 1 & 1 & 2 \\ 1 & x & 1 & -1 \\ 3 & 2 & x & 1 \\ 1 & 1 & 2x & 1 \end{vmatrix}$$

$$\begin{aligned} & (-1)^{\tau(1234)} a_{11} a_{22} a_{33} a_{44} + (-1)^{\tau(1243)} a_{11} a_{22} a_{34} a_{43} \\ &= x^3 + (-2x^3) = -x^3 \end{aligned}$$

故 x^3 的系数为 -1 .



行列式几何解释



不妨设 α_1 的长度(模)为 l , α_2 的长度(模)为 m , α_1 与 x 轴正向的夹角为 α , α_2 与 x 轴正向的夹角为 β , 于是, 如图 1-1 所示:

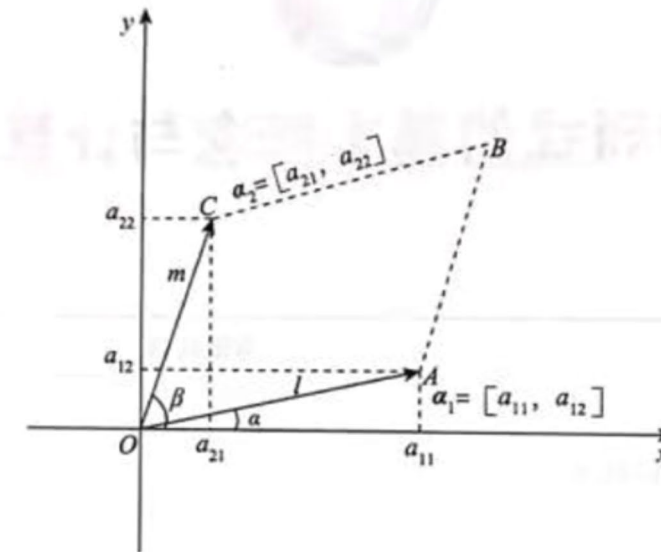


图 1-1

则

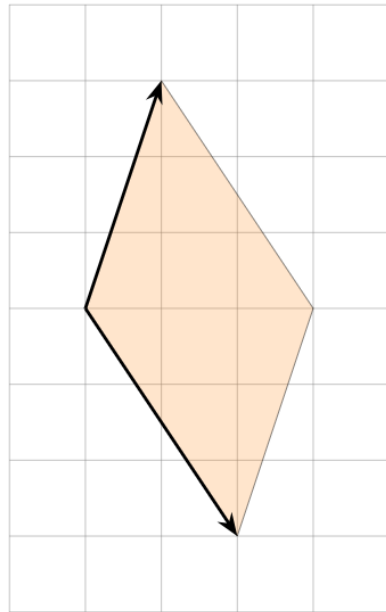
$$\begin{aligned} S_{\square OACB} &= l \cdot m \cdot \sin(\beta - \alpha) \\ &= l \cdot m (\sin \beta \cos \alpha - \cos \beta \sin \alpha) \\ &= l \cos \alpha \cdot m \sin \beta - l \sin \alpha \cdot m \cos \beta \\ &= a_{11}a_{22} - a_{12}a_{21}, \end{aligned}$$

于是 $\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21} = S_{\square OACB}.$

几何解释

[Determinants and Volumes \(gatech.edu\)](https://gatech.edu) 4.3 Determinants and Volumes

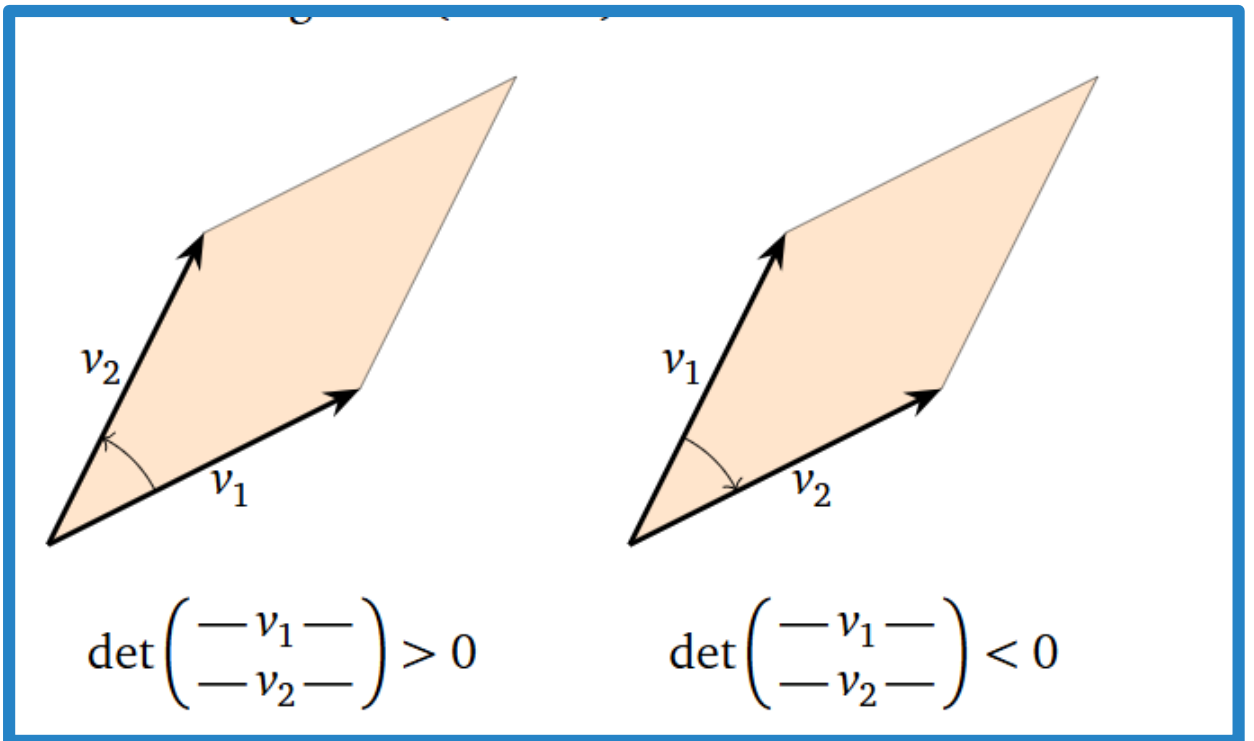
Find the area of the parallelogram with sides $(1, 3)$ and $(2, -3)$.

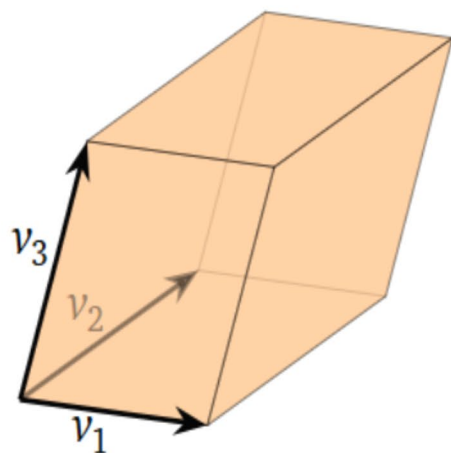


Solution

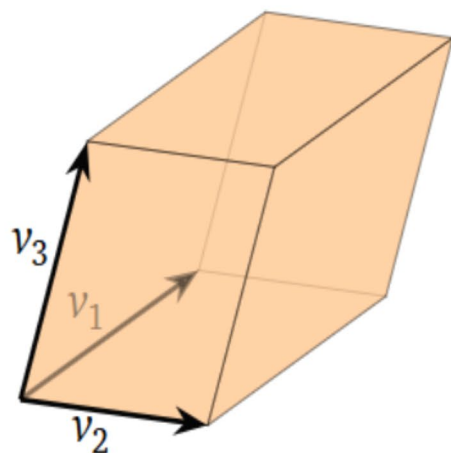
The area is

$$\left| \det \begin{pmatrix} 1 & 3 \\ 2 & -3 \end{pmatrix} \right| = |-3 - 6| = 9.$$

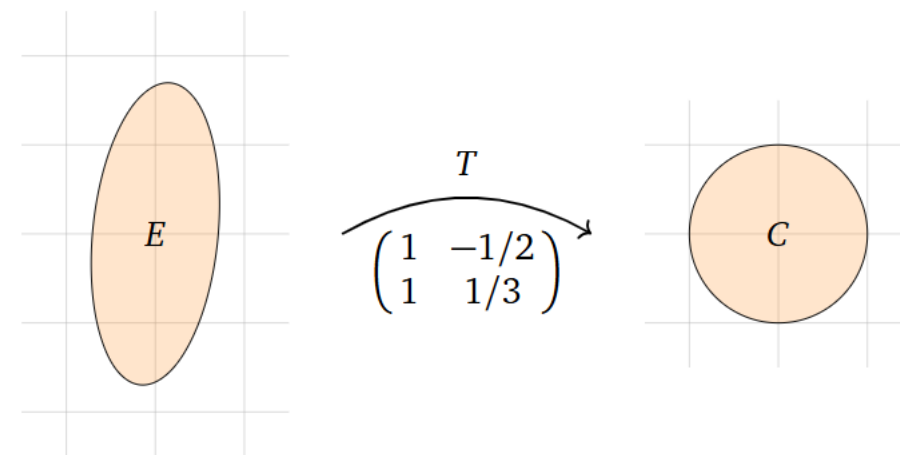




$$\det \begin{pmatrix} -v_1 - \\ -v_2 - \\ -v_3 - \end{pmatrix} > 0$$



$$\det \begin{pmatrix} -v_1 - \\ -v_2 - \\ -v_3 - \end{pmatrix} < 0$$



We compute the standard matrix A for T by evaluating on the standard coordinate vectors:

$$T \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad T \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -1/2 \\ 1/3 \end{pmatrix} \quad \Rightarrow \quad A = \begin{pmatrix} 1 & -1/2 \\ 1 & 1/3 \end{pmatrix}.$$

Therefore, T scales areas by a factor of $|\det(A)| = |\frac{1}{3} + \frac{1}{2}| = \frac{5}{6}$. The area of the unit circle is π , so

$$\pi = \text{vol}(C) = \text{vol}(T(E)) = |\det(A)| \cdot \text{vol}(E) = \frac{5}{6} \text{vol}(E),$$

and thus the area of E is $6\pi/5$.

课后思考：

$$\begin{cases} \alpha_{11}x_1 + \alpha_{12}x_2 + \dots + \alpha_{1n}x_n = b_1 \\ \alpha_{21}x_1 + \alpha_{22}x_2 + \dots + \alpha_{2n}x_n = b_2 \\ \dots\dots\dots \\ \alpha_{n1}x_1 + \alpha_{n2}x_2 + \dots + \alpha_{nn}x_n = b_n \end{cases}$$

系数矩阵，满足什么条件，经过初等行变换：数乘变换和倍加变换（不使用对换变换），可以将Gauss消元过程进行到底？



课后思考：

一个 n 行 n 列的矩阵，满足什么条件，经过经过初等行变换数乘变换和倍加变换（不使用对换变换），可以化为的行阶梯矩阵为上三角形矩阵？

对角线以下全为零的矩阵





作业：习题2.2: 2 (4) , 3, 5

- 排列
- n 阶行列式定义
- 行列式的性质
- 行列式的计算
- 行列式展开
- Cramer法则